

Azure & Render Deployment Fixes

This document outlines the fixes made to resolve deployment issues with both Azure App Service and Render.

Multi-Platform Database Configuration

The application is configured to work on **Render** (PostgreSQL via DATABASE_URL), **Azure App Service** (PostgreSQL via individual DB_* variables), and local development* (SQLite).

How It Works

```
# Multi-platform database configuration:
# - Priority 1: Render - Uses DATABASE_URL environment variable (PostgreSQL)
# - Priority 2: Azure - Uses individual DB_* environment variables (PostgreSQL)
# - Priority 3: Local - Falls back to SQLite for development
if os.environ.get('DATABASE_URL'):
    # Render configuration - use dj_database_url for PostgreSQL
    DATABASES = {
        "default": dj_database_url.config(
            default=os.environ.get('DATABASE_URL'),
            conn_max_age=600,
            ssl_require=not DEBUG,
        )
    }
elif os.environ.get('DB_ENGINE'):
    # Azure configuration - use individual DB_* environment variables for PostgreSQL
    DATABASES = {
        'default': {
            'ENGINE': os.environ.get('DB_ENGINE'),
            'NAME': os.environ.get('DB_NAME'),
            'USER': os.environ.get('DB_USER'),
            'PASSWORD': os.environ.get('DB_PASSWORD'),
            'HOST': os.environ.get('DB_HOST'),
            'PORT': os.environ.get('DB_PORT', '5432'),
            'OPTIONS': {
                'sslmode': 'require',
            },
        }
    }
else:
    # Local development fallback to SQLite
    DATABASES = {
        'default': {
            'ENGINE': 'django.db.backends.sqlite3',
            'NAME': BASE_DIR / 'db.sqlite3',
        }
    }
```

Platform Configuration

Platform	Configuration Method	Database Used
Render	<code>DATABASE_URL</code> env var	PostgreSQL via <code>dj_database_url</code>
Azure	Individual <code>DB_*</code> env vars	PostgreSQL via <code>psycopg2</code>
Local Dev	None	SQLite (local file)

Azure App Service Configuration

Required Environment Variables for Azure PostgreSQL

Add these environment variables in **Azure Portal** → **App Service** → **Configuration** → **Application settings**:

Variable	Value	Example
<code>DB_ENGINE</code>	<code>django.db.backends.postgresql</code>	<code>django.db.backends.postgresql</code>
<code>DB_NAME</code>	Your database name	<code>lensmaster_db</code>
<code>DB_USER</code>	Database username	<code>lensmaster_admin</code>
<code>DB_PASSWORD</code>	Database password	<code>your-secure-password</code>
<code>DB_HOST</code>	PostgreSQL server hostname	<code>lensmaster-server.postgres.database.azure.com</code>
<code>DB_PORT</code>	PostgreSQL port (default: 5432)	<code>5432</code>
<code>SECRET_KEY</code>	Django secret key	<code>your-django-secret-key</code>
<code>DEBUG</code>	Set to False in production	<code>False</code>
<code>ALLOWED_HOSTS</code>	Your Azure domain	<code>lensmasterpro-xxx.azurewebsites.net</code>

Optional Environment Variables

Variable	Purpose	Example
CLOUDINARY_CLOUD_NAME	Cloudinary cloud name	your-cloud-name
CLOUDINARY_API_KEY	Cloudinary API key	123456789012345
CLOUDINARY_API_SECRET	Cloudinary API secret	your-api-secret

Azure PostgreSQL Setup

1. **Create Azure Database for PostgreSQL** (Flexible Server recommended)
2. **Configure networking** to allow Azure services access
3. **Add environment variables** to App Service Configuration
4. **Run migrations** via SSH or deployment script:

```
bash
python manage.py migrate
python manage.py collectstatic --noinput
```

Issues Fixed

1. Database Configuration Error (Fixed)

Problem: App crashing with `django.core.exceptions.ImproperlyConfigured: settings.DATABASES is improperly configured. Please supply the ENGINE value.`

Root Cause: `dj_database_url.config()` returns an incomplete dict when `DATABASE_URL` is not set.

Solution: Check if `DATABASE_URL` exists before using `dj_database_url.config()`, otherwise fall back to SQLite.

2. DisallowedHost Error (Fixed - Latest)

Problem: `ERROR:django.security.DisallowedHost:Invalid HTTP_HOST header: '169.254.130.4:8000'`

Root Cause: Azure's internal health check uses IP `169.254.130.4` as the Host header.

Solution: Added Azure internal health check IPs to `ALLOWED_HOSTS`:

```
ALLOWED_HOSTS.extend([
    # ... existing hosts ...
    '169.254.130.4', # Azure health check IP
    '169.254.130.1', # Additional Azure internal IP
    '0.0.0.0',       # Catch-all for internal traffic
])
```

Files Changed

1. `lensmaster_pro/settings.py` - Fixed database configuration and ALLOWED_HOSTS
2. `AZURE_DEPLOYMENT_FIXES.md` - This documentation file